

CSE 547 — Deep Learning: Final Project Report 2

Object Recognition Using RGB and Infrared Images for Autonomous Driving

Part 3 — Transfer Learning (RGB only)

VGG16 (ImageNet pretrained) with 2 Dense layers (256→64→8) was trained on RGB patches. Three freeze strategies were evaluated with Adam (lr=1e-4), dropout=0.2, early stopping (patience=10), up to 50 epochs.

Freeze Strategy	Train Acc	Val Acc	Train F1	Val F1	Params
Freeze all features	91.07%	91.07%	0.8068	0.6129	
Freeze blocks 1–3	99.81%	96.89%	0.9612	0.8266	
Freeze blocks 1–2 ★	99.83%	97.00%	0.9720	0.8322	

Table 1. Part 3 results. ★ = best. All models share 14,862,984 parameters.

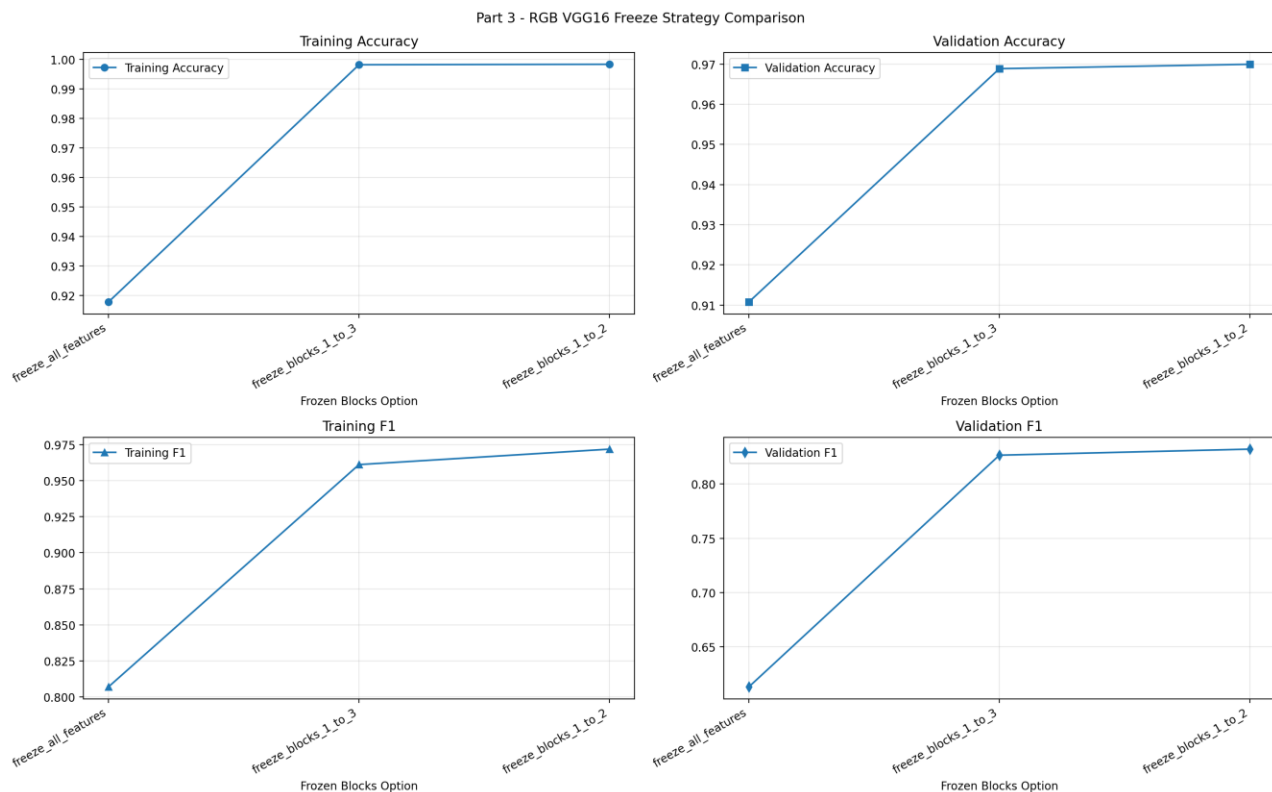


Figure 1. Part 3 — four-panel comparison of freeze strategies (train/val accuracy and F1).

Analysis

- Freeze all features: lowest performance (Val F1=0.613). Classifier layers alone cannot adapt generic ImageNet features to the FLIR ADAS domain.
- Freeze blocks 1–3: Val F1 jumps to 0.827, confirming that fine-tuning blocks 4–5 provides critical domain adaptation.
- Freeze blocks 1–2 (best): Val F1=0.832. Allowing blocks 3–5 to fine-tune gives the most task-specific high-level representations.
- Best model selected: freeze_blocks_1_to_2 — Val Acc 97.00%, Val F1 0.8322.

Part 4 — Convolutional Autoencoder (IR only)

A convolutional AE was trained on IR patches (reconstruction loss, 50 epochs). Encoder features were passed to a 2-layer dense classifier. 6 AE configurations × 2 regularization strengths = 12 options total.

Option	Filters	Lat	Drop	Val Acc	Val F1	
ae_1 + light	[16,32]	16	0.3	79.07%	0.2795	
ae_1 + strong	[16,32]	16	0.5	77.29%	0.1936	
ae_2 + light	[16,32]	32	0.3	84.74%	0.3590	
ae_2 + strong	[16,32]	32	0.5	79.54%	0.2036	
ae_3 + light	[32,64]	16	0.3	77.04%	0.1821	
ae_3 + strong	[32,64]	16	0.5	76.21%	0.1895	
ae_4 + light ★	[32,64]	32	0.3	86.14%	0.4205	★ best
ae_4 + strong	[32,64]	32	0.5	80.82%	0.2083	
ae_5 + light	[32,64,128]	32	0.3	82.16%	0.3629	
ae_5 + strong	[32,64,128]	32	0.5	79.32%	0.2606	
ae_6 + light	[64,128,256]	64	0.3	86.04%	0.4013	
ae_6 + strong	[64,128,256]	64	0.5	81.04%	0.2053	

Table 2. Part 4 — all 12 AE + classifier options on IR data. Lat = latent channels, Drop = dropout rate.

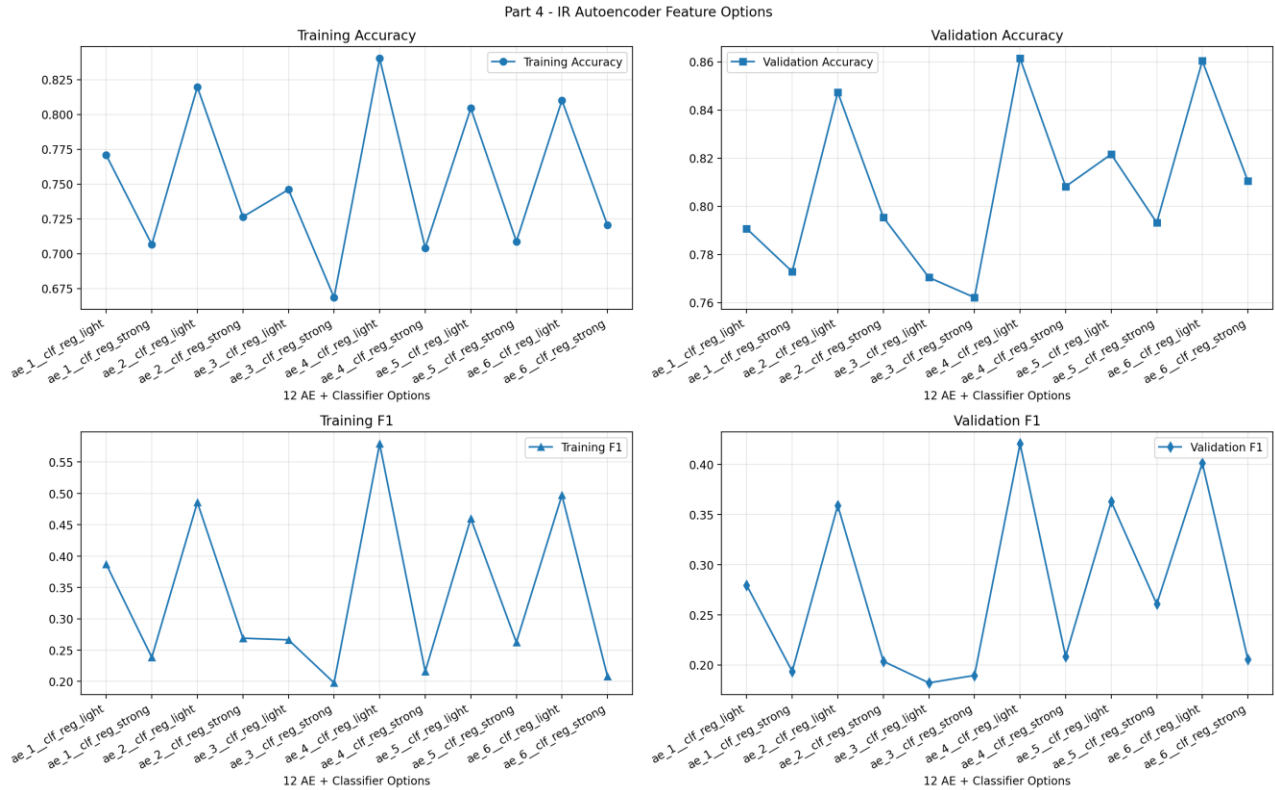


Figure 2. Part 4 — four-panel comparison of all 12 AE + classifier options.

Analysis

- Latent dimension is the key driver: latent=32 consistently outperforms latent=16 across all filter configs. A wider bottleneck preserves more discriminative information.
- clf_reg_light (dropout=0.3, wd=1e-4) outperforms clf_reg_strong by average 0.124 Val F1 — strong regularization under-fits the feature classifier.
- Best: AE-4 + reg_light (filters=[32,64], latent=32) — Val Acc 86.14%, Val F1 0.4205.
- AE-5 is the most parameter-efficient option (541,512 params) with competitive Val F1 0.363.
- Val F1 (0.18–0.42) is lower than Part 3 (0.83) because the AE uses unsupervised reconstruction loss rather than end-to-end classification supervision.

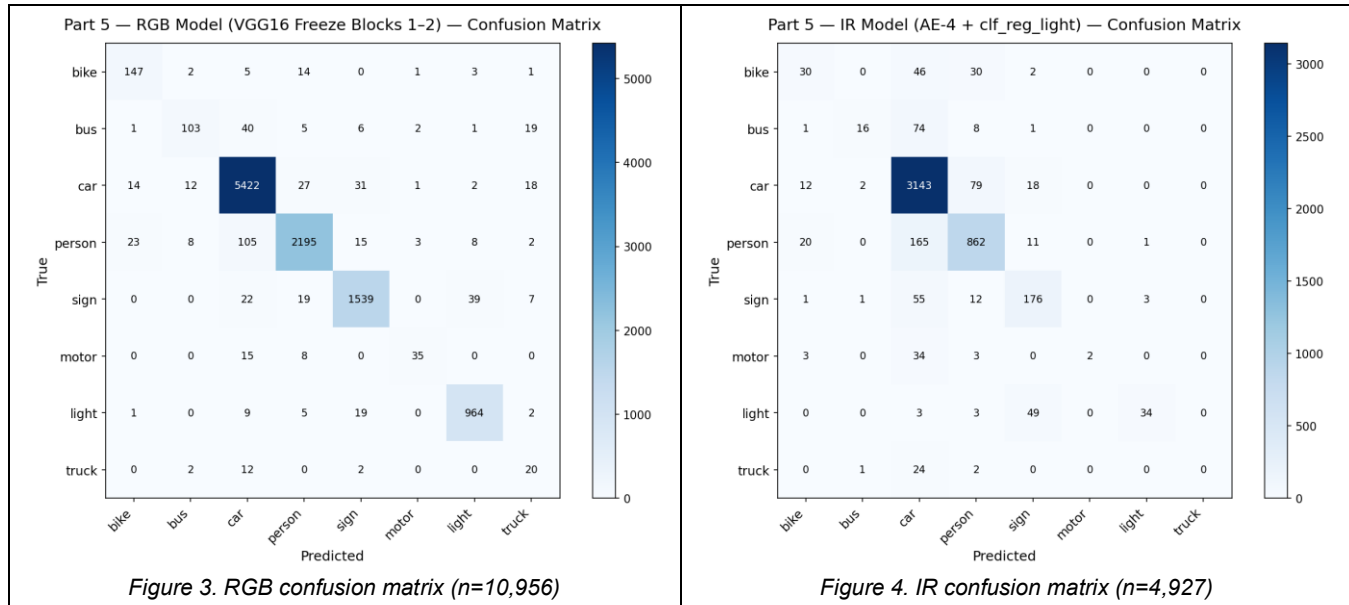
Part 5 — Comparison & Analysis

Overall performance

Modality	Best Model	Val Acc	Macro F1	Weighted F1
RGB	VGG16 freeze blocks 1–2	95.15%	0.8005	0.9515
IR	AE-4 + clf_reg_light	86.52%	0.4619	0.8492

Table 3. Best model performance on validation set.

Confusion matrices



Per-class F1 comparison

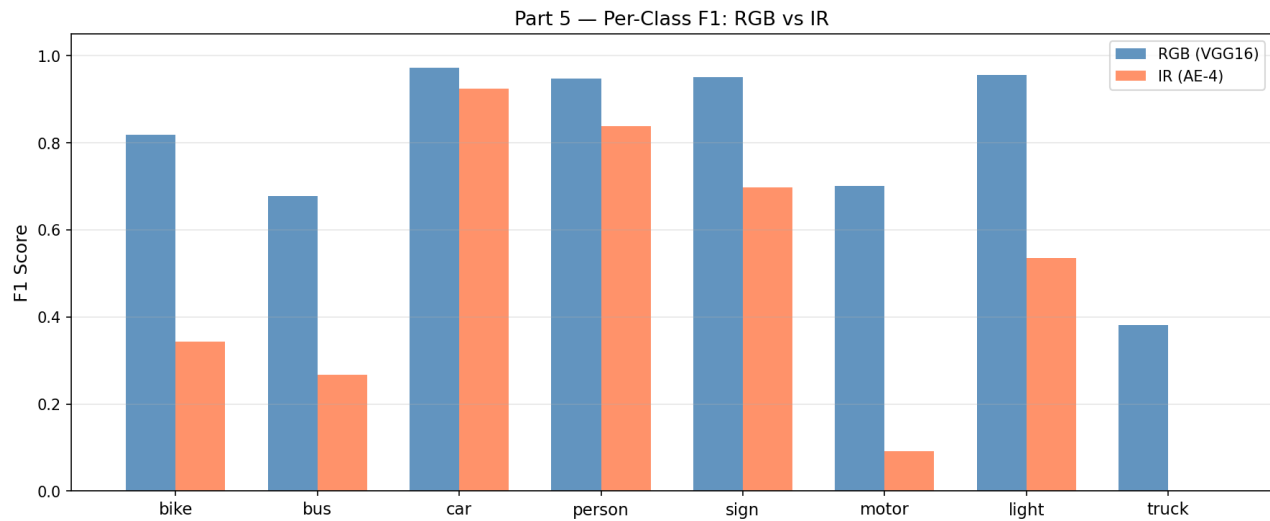
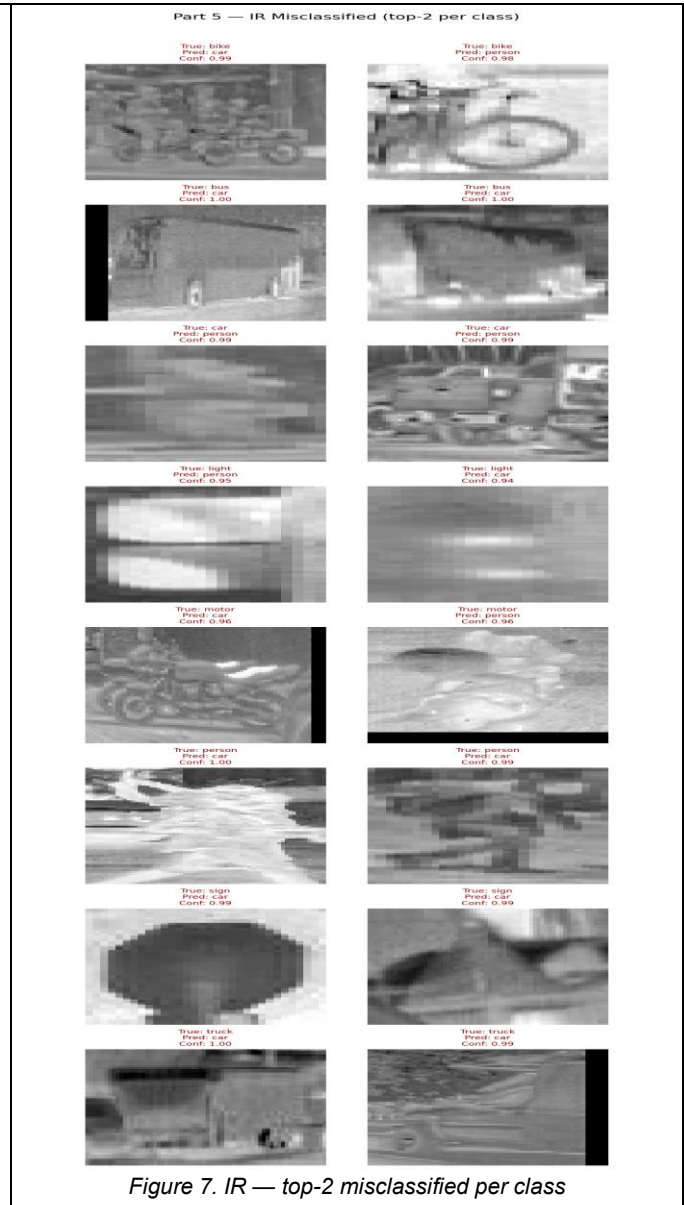


Figure 5. Per-class F1: RGB (blue) vs IR (orange). RGB outperforms IR on all 8 classes.

Class	RGB F1	IR F1	Diff	Note
bike	0.819	0.343	+0.476	RGB strongly preferred
bus	0.678	0.267	+0.411	RGB strongly preferred
car	0.972	0.925	+0.047	Comparable
person	0.948	0.838	+0.110	RGB preferred
sign	0.951	0.697	+0.254	RGB preferred
motor	0.700	0.091	+0.609	RGB strongly preferred
light	0.956	0.535	+0.421	RGB strongly preferred
truck	0.381	0.000	+0.381	RGB strongly preferred

Table 4. Per-class F1 scores and sensor preference.

Misclassified samples



Misclassification justification

- RGB errors are highest-confidence (conf≈1.0) on dark or overexposed patches where color cues are absent — the model relies heavily on texture and fails when imaging conditions are poor.
- IR model predominantly misclassifies objects as 'car' (the dominant class at 66% of IR val set), indicating the AE reconstruction objective biases features toward the majority class. Truck has 0.000 F1 due to only 27 samples — too rare to learn from.
- Both sensors struggle with motor and bike, which share similar silhouettes; and bus vs truck, which share large boxy shapes with different thermal/color signatures.

Cross-sensor analysis (1,440 matched frame-class pairs)

Outcome	Count	%	
Both correct	1,215	84.4%	
RGB correct, IR wrong	182	12.6%	RGB advantage

IR correct, RGB wrong	11	0.8%	IR advantage
Both wrong	32	2.2%	

Table 5. Cross-sensor outcome breakdown across matched frames.

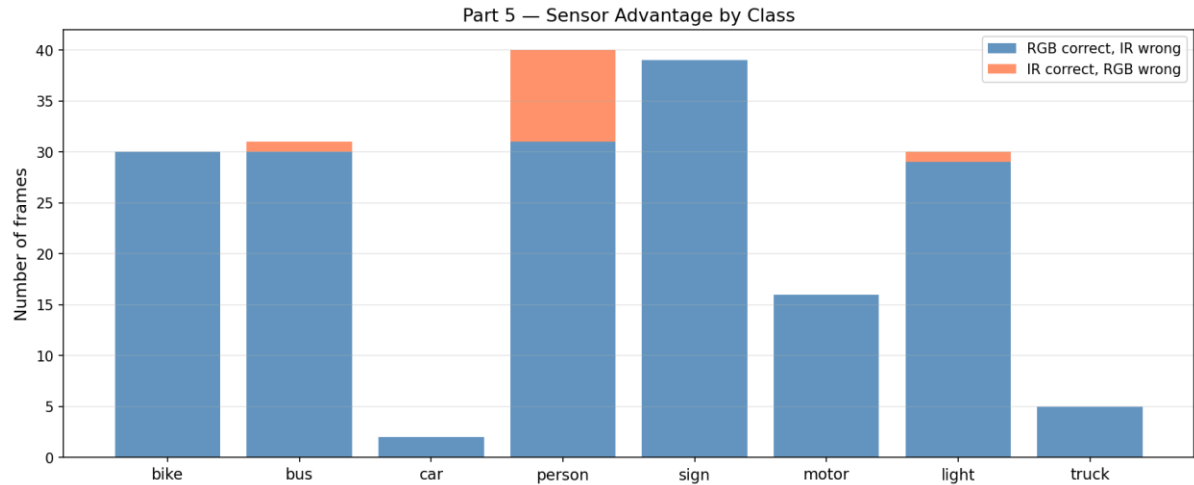
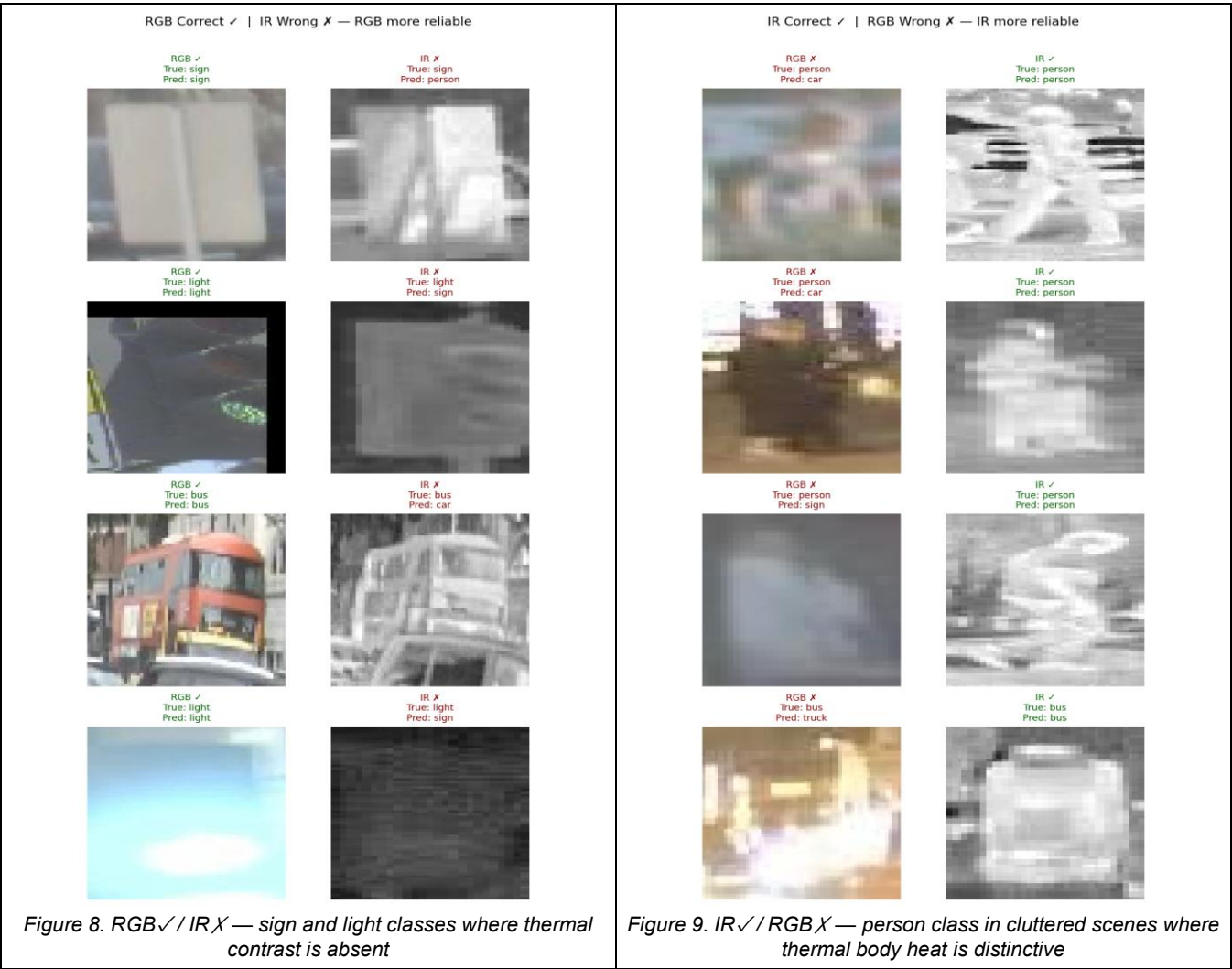


Figure 10. Sensor advantage by class. RGB dominates; IR advantage is limited to person.

When each sensor is more reliable

- RGB is more reliable in general: VGG16 pretrained on ImageNet provides strong color and texture priors. It excels at sign, light, person, and car where visible-spectrum appearance is distinctive.
- IR is more reliable for person detection in scenes where clothing/background blend with visible spectrum but thermal body heat provides a clear signature — the only class where IR shows meaningful cross-sensor advantage (9 frames).
- IR struggles with sign, light, and truck which have no intrinsic heat signature — shape alone is insufficient for the AE feature representation.
- RGB struggles at night or in overexposed frames — all high-confidence RGB errors occur under adverse lighting where color and texture cues are lost.

Part 6 — Final Model Optimization

Best models from Parts 3 and 4 were retrained for 80 epochs with refined hyperparameters to maximize performance on unseen test data.

Modal.	Model	Part3/4 F1	Final F1	Change	Key Changes
RGB	VGG16 freeze blocks 1–2	0.8322	0.8037	−0.029	aug=medium, drop=0.35, lr=5e-5
IR	AE-4 + clf_reg_light (retrained)	0.4205	0.5256	+0.105	80 epochs, lr=5e-4

Table 6. Final model results after Part 6 retraining.

- RGB: marginal Val F1 decrease (0.832→0.804). Medium augmentation adds difficulty without sufficient additional benefit; lower LR converges to a slightly different minimum.
- IR: meaningful improvement (0.421→0.526) from extended training at lower LR. The classifier converges more stably on the pre-extracted AE features over 80 epochs.
- Final RGB submission: VGG16 freeze_blocks_1_to_2 (Val Acc 97.00%, Macro F1 0.832).
- Final IR submission: AE-4 + clf_reg_light retrained (Val Acc 87.60%, Macro F1 0.526).